NE-Rank: A Novel Graph-based Keyphrase Extraction in Twitter

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***Abstract*—The massive growth of the micro-blogging service Twitter has shed the light on the challenging problem of summarizing a collection of large number of tweets. This paper attempts to extract topical keyphrases that would represent topics in tweets. Due to the informality, noise, and short length of tweets, such research is nontrivial. We tackle such challenges with extensive preprocessing approach. Followed by, introduction of new features that improve topical keyphrase extraction in Twitter. We start by proposing a novel unsupervised graph- based keyword ranking method, called NE-Rank, that considers word weights in addition to edge weights when calculating the ranking. Then we introduce a new approach of leveraging hashtags when extracting keyphrases. We have conducted a set of experiments showing the potential of both approaches with 16% to 39% improvement for NE-Rank and 20% improvement for hashtag enhanced extraction.**

***Keywords*-Keyphrase Extraction, Graph-based Ranking, Hashtag, Twitter, PageRank, TextRank, NE-Rank;**

I. INTORDUCTION

Micro-blogs have been recently attracting people to express their opinion and socialize with others. One of the most popular micro-blog sites is Twitter[1] [1][2][3]. People blog about anything they would want to share. Not only people use the giant micro-blog service, but also news agencies and corporations started to use it to disseminate news and advertisement. The growth of Twitter users created a huge number of generated posts every second, called tweets. These tweets can contain text, usernames, hashtags (a Twitter convention to tag tweets with keywords), and URL links. Even though every tweet has to be less than 140 characters, it did not limit the popularity of using Twitter. In fact, Twitter has grown exponentially with billions of users' generated posts. These posts cover an immense range of topics and interests resulting in a vast volume of text. This growth presented a great need of studying and analyzing contents of tweets in hope of summarizing the huge collection of posts. This need had led to raise the following research questions: How can we summarize a collection of tweets? How to organize the summary in respect to topics? Can we summarize tweets using keyphrase extraction? How to improve it further?

In this paper, we address these questions by proposing an enhanced PageRank[4] inspired graph-based algorithm for unsupervised keyphrase extraction designed specifically for Twitter. We investigate the possibility of producing keyphrases capable of summarizing tweets. Moreover, we identify topics in tweets to organize keyphrases with respect to their topics. Keyphrases are phrases that can signify the document they appear in. Extracting keyphrases generally can be tackled in two approaches. First, supervised approach that uses a training model and treats the task as a classification problem[5][6][7]. In this technique, human labeled dataset is needed to train a model to correctly identify keyphrases. This approach has shortcomings due to the fact that it depends on the trained model and the domain it is trained on. This drawback means that such an approach will struggle to scale to new domains [8]. In an environment such as Twitter with diverse domains and topics, it is impractical to consider this approach. The second approach is by using an unsupervised method where no human labeled dataset is necessary which is suitable for a dynamic text as in micro-blogs. In this paper, we use an unsupervised approach using graph-based models.

Although extracting keyphrases has been tested on documents in news, scientific papers where a formal usage of languages is generally assumed [9][10][11], extracting keyphrases from Twitter is still in its experimental stages [12][13]. In a micro-blog as in Twitter, the documents are much shorter with an informal language usage. Attempting to extract keyphrases from tweets that embody conversational language style is not a trivial task. Tweets are very noisy where they include a handful of symbols and emoticons which make it a challenge to extract keyphrases or mine the textual content of the posts collection. Tweets are commonly written by users without proper grammar or sentence structure. They, also, could be taken from news headlines without forming a clear language structure which makes using natural language processing methods a difficult task.

This paper improves the state-of-the-art methods for extracting topical keyphrases from Twitter [12] by providing the following contributions: First, a new graph-based approach that uses frequency measures to enhance the ranking formula. Second, a new approach of leveraging hashtag titles to find hidden keyphrases that do not appear in the corpus.

The paper is organized as follow: A discussion of the related work is in Section II. The proposed approach will be explained in Section III. Section IV will describe the

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data and experimental results. The paper conclusion is in Section V.

II. Related Work

Our research is based on graph-based unsupervised ranking algorithm TextRank [9]. TextRank was inspired by PageRank [4] by using the ranking algorithm for text. LexRank is another lexical graph-based approach used for identifying the most important sentences for summarization purposes [14]. A variant of TextRank had been proposed to incorporate topics when extracting keyphrases. A topic decomposition framework was discussed by Liu et al. [11] In their work, they included multiple random walks for topics instead of the traditional single random walk through the graph. However, their algorithm only gives higher probability for each topic in a separate graph without taking the context of topics in the formula that measures the ranking. The work in [12] is very close to our research. They extracted topical keyphrases from a collection of tweets. Our research is different in the following: We are giving hashtag keywords that appear in the hashtag title as an attached string a higher weight after splitting them into separate tokens; and we are using TextRank algorithm with an enhanced formula that combines additional weight measures for the nodes and edges of the graph. Another unsupervised keyword extraction is shown in [13]. In their work they extracted keywords from tweets to tag users with their interest . They compared TextRank with $TFIDF$(Term Frequency - Inverse Document Frequency) to rank keywords for each user. Our work is different than [13] in that we extract keyphrase while they only extract keywords. Moreover, we extract topical keyphrase to summarize topics within a collection of tweets instead of extracting keywords for tagging user with their keywords. Wan and Xiao used the neighboring documents to extend TextRank graph that showed to improve the results when extracting keyphrases from a single document [15].

A different line of research that is relevant to keyphrase extraction is the supervised approach. Hulth used a machine learning approach to extract keywords by using linguistic features to train her model [10]. KEA [6] is another supervised system that uses Naïve Bayes classifier that considers $TFIDF$ and the position of the word in the document to be significant features in classifying a word as a correct keyword. An improved KEA++ [7], extends the previous algorithm by using semantic information through a controlled vocabulary, for example, Wikipedia.

III. Proposed Approach

In this section, the overview of the research approach will be discussed. We find latent topics in text and cluster tweets based on topics found. Then, the graph-based ranking algorithms and a leveraging hashtags titles approach will be illustrated. Finally, generating a keyphrase candidate list and ranking of keyphrases will be explained.

A. System Overview

Twitter has a great volume of tweets encompassing so many topics. First step in our work is to identify topics in the dataset to extract topical keyphrases. Furthermore, we followed a well-known approach for keyphrase extraction which is to rank keywords, generate candidate keyphrases, and rank keyphrases as a post-processing step [12][9][11]. We also extracted hashtags for further processing and weighting which we are investigating if it might affect the ranking. The next sections will discuss topics extraction, keyword ranking algorithms, hashtags extraction, keyphrase candidate generation, and finally keyphrase ranking.

B. Topic Extraction

Given a collection of tweets $\mathcal{T}_i = \{\tau_1, \tau_2, ..., \tau_n\}$, we assume that the collection $\mathcal{T}_i$ would include K multiple topics. To be able to extract topical keyphrases, we need to identify topics in the collection $\mathcal{T}_i$ then extract keyphrases for each topic K. We used a topic model to recognize the topics distribution over the tweets. The model is a well-known unsupervised technique used to detect latent topical information called **Latent Dirichlet Allocation (LDA)** [16]. The model assumes a bag of words approach. Moreover, every document is treated as a vector of words frequencies. Therefore, the model assumes that every document is modeled as a probability distribution over number of topics, and every topic is modeled as a probability distribution over words. LDA uses a generative process for every document:

1- Choose a topic for every document from the topic distribution.
2- From the words distribution of the topic, sample a word from the chosen topic.
3- Repeat the process for every word within the document.

LDA models a generative method of documents as stochastic process where words are chosen by sampling some discrete probability distributions. Formally, every document is represented with a multinomial distribution over topics as θ. Also, every topic is represented as a multinomial distribution over words as ϕ. Both θ and ϕ have dirichlet prior parameters α and β respectively. For every word in a document, there is topic z sampled from the multinomial distribution θ. Consequently, a word w is sampled from the multinomial distribution ϕ. To form a document, this generative process is repeated for every word in the document.

Topic models are known to have difficulties when used with short text documents [17][18][12]. To avoid the problem of short length in text, we followed an approach similar to the work done by Hu et al. for expanding tweets[18]. Each tweet is expanded by querying a search engine using the tweet text as a query. Then, the top 5 search results snippets are retrieved and the top 10 $TFIDF$ terms are computed. These top 10 terms are inserted to the tweet before using the topic model LDA. Note

that such expansion approach is only used for extracting topics. For extracting keyphrases, normal tweets are used.

LDA assumes a K number of topics to be fixed. We empirically experimented with {10, 20, 30, 50, 100} with best topics achieved with K=30. Moreover, since tweets are of short length, we assumed that every tweet is about a single topic. Therefore, we used the highest topic distribution of the tweet to be the topic assigned to it. Then, we aggregate all tweets of the same topic to be a separate subdataset. The tweets collection $\mathcal{T}_i$ is the union of all topical subdatsets $\mathcal{T}_i = \{\tau_{k1}, \tau_{k2}, ..., \tau_{kn}\}$ where τ_k is a subdataset that belongs to topic k. Therefore, we run every τ_k independently to extract topical keyphrases, and call them topical dataset.

C. Graph-based Keywords Ranking

Graph-based ranking algorithms are inspired by Google's PageRank[4]. In PageRank, the goal is to rank webpages by measuring the importance of a webpage according to the entire web (graph) influence. Formally, let $G = (V, E)$ be a graph with directed edges E and a set of vertices V. Let V_j be a set of vertices that direct to V_i, and O(V_j) is the outdegree of the node V_j. PageRank calculates the rank of V_i as follows:

$$R(V_i) = (1-d) + d \cdot \sum_{j:V_j \to V_i} \frac{1}{|O(V_j)|} R(V_j) \quad (1)$$

d is the damping factor which is the probability of jumping from a node to the next node. The value of the damping factor is usually set to 0.85[4][9] which is what we used in our experiments. $(1-d)$ is the probability of jumping to a new node. The formula assumes unweighted edges for the graph. PageRank can be used with undirected graphs, as well, where the in-degree of the vertex is equal to the out-degree of the vertex, which is what we used in our implementation. First, the nodes are set to a uniform default value. Second, the ranking algorithm runs on the graph until a convergence is reached. The convergence is the difference value between a current iteration and a previous iteration being less than a specific threshold. In our implementation we set the threshold to 0.0001.

To apply graph-based ranking algorithms on text, we build a graph for text. Every vertex represents a word (we only focused on nouns and adjectives), and the edge represents the co-occurrence value between any two words within a specific window size. In our experiment we used a window of size 2. The output of such algorithms is a list of ranked keywords that represent the document.

A weighted graph has been proposed by Mihalcea & Tarau that takes into account the number of co-occurrences between two words as the weight of the edge [9]. In this paper, we will refer to such approach as *TextRank* which calculates the ranks as the following:

$$R(V_i) = (1-d) + d \cdot \sum_{j:V_j \to V_i} \frac{w_{ji}}{\sum_{k:V_j} w_{jk}} R(V_j) \quad (2)$$

where w_{ji} is the weight of the edge from the previous vertex V_j to the current vertex V_i, and $\sum_{(k:V_j)} w_{jk}$ is the summation of all edge weights in the previous node V_j. Including edge weights give the ranking algorithm the ability to recognize the strength of the co-occurrence relation.

Both PageRank and TextRank do not take a weight for the node to calculate the ranking. *Therefore, we introduce a novel ranking formula that takes into account both edge and node weights*. PageRank was originally used for the Web to measure websites' importance using only the link structure of the graph. However, when used in text to represent a lexical graph, it is possible to include a weight for the words that will measure the ranking more accurately instead of only relaying on the co-occurrence. Unlike TextRank where the algorithm only focuses on the relationship between words, we develop our approach to also consider words weight as well. We integrate the nodes' weight into the formula where it affect the probability of jumping from a node to the next (the damping factor). Instead of having a "random surfer model" with equal probability of jumping to the next node, we give nodes with higher weights a priority. This will result in a more natural ranking of text instead of only focusing on the co-occurrence relationship since it considers the weight of words too. For example, if two unimportant words are co-occurring frequently, then they will be ranked higher in the list regardless of their importance. Moreover, if an important word happens to not occur with a specific set of words frequently, then the important word will not be recognized as important in the ranking. Using the weight of nodes with TextRank will result on a node-edge weighting approach as in this new formula that we call NE-Rank (Node and Edge Rank):

$$R(V_i) = (1-d) \cdot W(V_i) + d \cdot W(V_i) \cdot \sum_{j:V_j \to V_i} \frac{w_{ji}}{\sum_{k:V_j} w_{jk}} R(V_j) \quad (3)$$

where $W(V_i)$ is the weight of the current node. In our implementation, we have experimented with normalized $TFIDF$ to be the weight of the node which produced interesting results. For measuring weights with $TFIDF$, we define the statistical measure as in [2]:

$$W(V_i)_{TFIDF} = tf(V_i) \cdot log_2 \frac{N}{df(V_i)} \quad (4)$$

Due to the short size of tweets, Hybrid $TFIDF$ will measure $tf(V_i)$ as the term frequency on the whole collection of topical tweets. N is the number of documents (tweets) in the topical set, and $df(V_i)$ is the number of documents that contain the term V_i. All of the ranking algorithms run on every topical dataset separately to produce the ranking of keywords within a specific topic. Finally, for generating candidate keyphrases, we select the top T keywords then look for adjacent top keywords to form a candidate keyphrase list for a specific topic.

D. Hashtags Titles

Hashtags are a user-convention adopted by Twitter to aid in indexing, archiving, tagging tweets that contain

them. A user can prefix any word with a pound sign "#" to make it a hashtag to mark a keyword. It can also precede a phrase which will appear as detached string of words, for instance "#GlobalWarming" or "#OccupyWallStreet". The usage of hashtags in tweets is shown to be wide and common [19] [20]. Hashtags are considered as a metadata addition to tweets that can add some semantical information. The emergence of hashtags and why people use them is still a research matter and our understanding is still limited. One study explores if hashtags can have a dual role of tagging keywords and identifying virtual community [20]. Another study is measuring sentiment in hashtags [19]. The usage of hashtags as keywords annotation makes them of a very interest to our work. Such annotation can be very helpful in a task such as keyphrase extraction. Most work in keyphrase extraction simply ignores hashtags which is a valuable annotation from users. Therefore, *in this paper, we propose a novel approach of leveraging hashtag titles in the process of extracting keyphrases*.

Consider the following example of an imaginable tweet: "*I just helped by donating money #VoteObama*." Clearly, important keywords are "Vote" and "Obama" that need to be utilized instead of being ignored. We have utilized the keywords that appear in a hashtag as the following: First, we extract all the hashtags from every topical dataset. Second, every hashtag title is split into keywords if it encompasses more than one word, and then we boost the weight of these keywords to show their importance. Formally, let $\mathcal{H}_i = \{h_1, h_2, ..., h_n\}$ be a set of hashtags in the topical dataset. For every h_i that appear to encompasses more than one word in its title, for example "#BarackObama", they will be split into two keywords as in "Barack" and "Obama". We develop a tool which segments hashtag string to tokens using an English dictionary with frequencies. Such an approach produces minor number of wrong results that were corrected manually. Moreover, these keywords signify important tags to the topic they belong to. We, then, boost the ranking of such keywords to use them in extracting important keyphrases.

To boost the weights of hashtags that are important to the topic, we designed a boosting strategy using the graph features. In the lexical graph, all nodes with hashtags are identified. Then, we record the number of the in-degree edges to the hashtag node that signify the importance of a hashtag. Hashtags were boosted based on the percentage of the in-degree value. For example, if a hashtag node has 10 edges, then the hashtag will be boosted 10%. We focused on boosting only hashtags that are relevant to the topic. To determine the relevancy of a hashtag, we measured the top-10 $TFIDF$ words in the topical tweets. Then, we only consider the hashtag to be relevant if it overlaps with one of the top 10 words. In Section IV, we will show how such approach of boosting hashtags was very rewarding. Moreover, the relevancy of hashtags can be measured using a topical ratio that helps in focusing on only topical relevant hashtags. By using this ratio with the percentage of the in-degree for the hashtag node, we can only boost relevant hashtags as follows:

$$boosting = \frac{|h_i|}{|H|} \cdot p(edges) \tag{5}$$

$$hashtagrank = NE\text{-}Rank(h_i) + boosting \tag{6}$$

where $|h_i|$ is the number of the hashtag appearances in the topical dataset, and $|H|$ is the number of the hashtags appearances in the general corpus. $p(edges)$ is the percentage of the in-degree edges for the hashtag node.

E. Candidate Keyphrase Generation

After ranking keywords, we generate a candidate keyphrase list of all possible keyphrases for each topic. In this section, we describe how we generated such a list. First, all the ranked keywords were ordered in a descending order. The top T keywords were used to generate keyphrases. Second, the positions of the top keywords were searched in the corpus to locate where top keywords occurred. Third, we checked the words appearing right after the position of the top keyword. If a word in the next position happens to be a top keyword as well, we designate the phrase of both keywords as a *candidate keyphrase*. Keyphrases are generally known to be noun phrases when annotated by humans [10][11][15]. Therefore, we try to focus on noun phrases. Even though noun phrases can be longer than two words, we focus on extracting short phrases without using linguistic knowledge as in noun phrase boundary detection methods. A large list was produced of candidate keyphrases containing from 721 to 1302 phrase for every topic. In the next section, we illustrate strategies for ranking phrases and identifying the top keyphrases that represent each topical tweet collection.

F. Keyphrase Ranking

We explain different approaches for filtering and ranking candidate keyphrases list. To get the top keyphrases that represent the topical dataset they belong to, we need to rank them and filter out any unwanted phrases. Since the generation process of keyphrases was done automatically, some phrases will be incomprehensible and won't fit together to be a phrase. For example, if an important keyword happens to occur right before the end of a sentence boundary, the next word will appear in the next sentence which means that they cannot be considered as a keyphrase. Therefore, we filtered out any keyphrases that cross syntactical boundary. Moreover, to ensure the two words are a good fit for a keyphrase, keyphrases occurring less than 5 times were filtered out to improve phresness. Any duplicate keyphrases were removed. In section 4, we will describe the preprocessing step in details, for example: stopwords removal. In order to recognize the top keyphrases, a ranking mechanism is needed.

Let $k_i = \{R_1, R_2\}$ be a keyphrase and R_i is the keywords that constitute the keyphrase. Most keyphrase extraction research simply applies the summation of keywords score to rank the keyphrase as follows:

$$R(k) = \sum_{R \in k} R(V_i) \tag{7}$$

where R(V_i) is the rank of the keywords [9][15][11]. However, this approach will be strongly biased toward strong keywords but not necessarily strong keyphrase. For instance, if only a single keyword is very strong and the other keyword is at the bottom of the list, it would still incorrectly rank keyphrase higher than a better keyphrase that does not contain a dominantly strong keyword. To measure the strength of the keyphrase another approaches that take the product of both keywords into account will add some balance to the ranking of keyhprases, as in the following:

$$R(k) = \sum_{R \in k} log R(V_i) \tag{8}$$

After experimenting with both approaches, we found the latter to produce better results. The final length of keyphrase for each topic ranged from 51 to 93 keyphrases. With some minor numbers of irrelevant keyphrases, the proposed approach was able to produce good topical keyphrases that seemed relevant to the topic they belong to. Hashtags boosted keywords appeared in the top list of keyphrases frequently. In particularly, hashtags with a composite title that has been split into separate keywords were very strong keyphrases that appeared in top 10 in multiple topics. We suspect that some topics have a heavy usage of hashtags compared to others that either do not have as many or they are not relevant to the topic and do not have a strong score. Finally, extracting keyphrases appearing as hashtags have improved extracting keyphrases that represent the topic.

IV. Experiment

Measuring the effectiveness of our approach for extracting keyphrases from a micro-blog medium has its own difficulties and challenges. NLP of-the-shelf packages for text processing such as Part of Speech POS taggers cannot be directly applied to tweets without extensive preprocessing approach. Therefore, in our experiment we tried to pay more attention to the preprocessing step.

In this section, a description of the experiment will be provided. First, the dataset will be explained, and the preprocessing steps will be discussed. Second, the evaluation metrics will be illustrated. Third, the experimental results will be discussed.

A. Dataset and Preprocessing

To perform an experiment on ranking keywords and extracting keyphrases, we gathered tweets for a period of three months. From December 2011, January, and February 2012, we gathered 31,227 tweets consisting of 244,139 tokens. The collection has more than 4,079 hashtags with a total frequency of 40,674 appearances.

Twitter is a micro-blog where people generally write in a conversational style. Tweets are known to be very noisy for any text mining task. They contain a number of symbols and smileys that make using standard preprocessing steps to be ineffective. Tweets usually contain misspelled or abbreviated words. They also often lack a proper use of capitalization or punctuation. All these facts make it very challenging to perform a keyphrase extraction experiment. In our experiment, first we filter out all non-Latin tweets by using regular expression. Then, we filtered out any non-English tweets by counting the words of English that match an English dictionary within every tweet to be more than the avrage length. Any URL links appearing in tweets were removed since we are only focusing on the textual body of the message. Even though emoticons and smileys can be helpful in other areas of research, as in sentiment analysis, they might affect the POS tagger accuracy, and therefore they were removed.Additionally, Twitter usernames starting with "@" symbol and the re-tweet string "RT" were removed. Moreover, tweets contain a number of Internet slangs that are mostly Out of Vocabulary OOV. Therefore, we tried to transform these slangs and abbreviation into standard English[2]. In more details, we used an extensive dictionary for Internet slangs and abbreviation consisting of 5340 words and phrases to transform the tweets collection into normal English by using a lookup table, for instance, abbreviations as in "b/c" for "because" or "b/w" for "between" were correctly substituted with their formal words. Additionally, some abbreviation affects the grammatical structure and would reduce the POS tagger accuracy, for example "I'm" instead of "I am" or "wl" for "will". Moreover, orthographical changes as in "2night" for "tonight" were corrected using the dictionary.

After an effort of transforming the tweets into standard English, we used a POS tagger to only focus on nouns and adjectives. POS has known approach: each word would be assigned with its common tag, and any OOV words will be assigned with the mostly used tag [21]. Since tweets contain a large number of OOV its accuracy will not be perfect. Also, the misuse or lack of use of punctuation and capitalization will affect the POS tagger. However, we decided to use it since we are not focusing on accurately tagging every word but only to limit the selection to nouns or adjectives.

Subsequently, we removed all common stopwords since they do not constitute a keyphrase. Moreover, we stemmed all words using Porter stemmer[3], and saved the position of stemmed words so they can be mapped back to their original form when the results are presented.

For extracting topics with LDA, we used Gibbs sampling with 500 iterations. We chose to extract 30 topics after experimenting with number of options. Moreover, since we assign every tweet with its highest topic, we aggregate all tweets for each topic in a separate dataset that is then passed to the graph-based ranking algorithms. For evaluation, we picked the best five topics to evaluate their results. In the next sections, we will discuss the evaluation of the test dataset that we used for evaluating keywords weighting and keyphrase extraction techniques.

[2]http://www.noslang.com/dictionary/full/

[3]http://tartarus.org/ martin/PorterStemmer/

Table I
TOP 10 KEYWORDS

	Precision	Bpref
PageRank	0.54	0.750
TextRank	0.68	0.754
NE-Rank	0.76	0.857

Table II
TOP 25 KEYWORDS

	Precision	Bpref
PageRank	0.36	0.623
TextRank	0.43	0.663
NE-Rank	0.51	0.686

Table III
TOP 50 KEYWORDS

	Precision	Bpref
PageRank	0.25	0.547
TextRank	0.28	0.560
NE-Rank	0.34	0.566

B. Evaluation Metrics

The common approach to evaluate keyphrase extraction is to compare it against a golden standard dataset by using a standard evaluation metrics as in precision and recall. Although a golden human annotated dataset does exist in forms of news or research papers with annotation, such dataset does not exist for Twitter. The lack of such human annotated dataset made the evaluation of such research a nontrivial task. As an alternative, we designed our evaluation by comparing against results obtained from a search engine.

To compensate the lack of golden labels to compare against, we have empirically designed an evaluation approach that utilized a search engine. The basic idea is to search the Web for each topic and generate labels that will be used as golden labels. For each topic, we retrieved the top 5 topical terms from LDA and send those words as query to Google. Top 10 snippets returned from the search engine were used. We only used the titles and description of each snippet. Then, after tokenization, we measured the term frequency among all documents retrieved from Google to generate a golden list. Given that the goal of extracting keyphrases from Twitter is to summarize the collection, our focus must be on the quality of the top ranked keyphrases. Therefore, we focus on the top precision with top 10, 25, and 50. Precision is measured as the following:

$$Precision = \frac{K_{correct}}{K_{extracted}} \tag{9}$$

where $K_{correct}$ is the number of correctly extracted keywords, and $K_{extracted}$ is the total number of extracted keywords. Even though precision measures how precise the extracted result is, it does not differentiate in the ranking of the extracted top result. Therefore, we used another metric that measure the ranking within the top extracted keywords. The metric is Binary Preference Measure [22]. Bpref is a great choice to measure the performance of the system with the order of the top extracted keywords taken into account. Bpref is measured as the following:

$$Bpref = \frac{1}{R}\sum_{r \in R} 1 - \frac{|\text{n } \textit{ranked higher than } \text{r}|}{M} \tag{10}$$

where R is the correct keywords within M extracted keywords in a method, and where r is a correct keyword and n is incorrect keyword.

C. Experimental Results

We will discuss the experimental results. First, we evaluate keywords ranking with graph-based methods. Second, we evaluate the extracted keyphrases and whether using hashtags have yielded any advantages or not.

To evaluate the performance of ranking keywords, we conducted a performance measurement using precision and Bpref. Now, we discuss the evaluation of all different ranking algorithms. We compared all three algorithms which are: PageRank (unweighted edges), TextRank (weighted edges only), and NE-Rank (weighted edges and nodes). Results are shown in Tables I, II, and III by measuring the precision and Bpref for top 10, 25, and 50 respectfully. When comparing TextRank to PageRank, we can see that incorporating edge weights for the strength of co-occurrence have yielded an avrage of 18% improvement for all percentages of top 10, 25, and 50 in precision. In NE-Rank we included both edges and nodes. From tables I, II, and III, we can see that NE-Rank clearly outperformed both PageRank and TextRank. The improvement over TextRank is around 16% in precision for all percentages and an average of 39% over PageRank. Using weights for words to enhance the ranking have showed better ranking and improved the graph-based ranking algorithm. The idea behind NE-Rank and why it improves the ranking is that relying only on the co-occurrence relationship is not enough. After our experiment, we showed that using the weight of words for nodes can improve the ranking when using graph-based methods. As shown in Figure 1, the precision of the extracted top keywords is better than all others. Moreover, the order of the top keywords was shown to be better by using the Bpref measure.

To evaluate keyphrases, since we found that NE-Rank performed best among others, we decided to choose it as the ranking algorithm for keyphrase extraction. After ranking keywords and generating keyphrases, we ranked all keyphrases one time with boosting hashtags and another without boosting them. Moreover, we investigated the possibility of increasing the performance through boosting hashtag title words. We can see in Table V that incorporating hashtags do improve results with precision by 20%. Also it adds higher quality and more relevant keyphrases to the final top results. The reason why hashtags are important is the fact that people, most of the time, use them on Twitter to signify topicality of the tweet they appear at. Leveraging hashtag titles to extract

Table IV
TOP 10 KEYPHRASES OF FIVE TOPICS (## REPRESENT A WORD SPLIT FROM A HASHTAG TITLE)

Topic 8	Topic 10	Topic 17	Topic 25	Topic30
##sandusky ##hearing [4]	saif #gaddafi	super bowl	#annahar lebanon	##occupy ##seattle
waives hearing	##col ##gaddafi	win super	#annahar #beirut	##seattle protestors
jerry sandusky	higgs boson	giants win	#lebanon #annahar	occupy protest
#sandusky waiving	#libya #col	bowl xlvi	annahar #syria	seattle #d12
sandusky waives	##zintan ##brigade	england #patriots	beirut lebanon	##media #politics
##psu ##charges	##cameron ##legacy	bowl tweets	lebanon syria	##social ##media
preliminary hearing	##gaddafi #ntc	ny giants	syria security	flash bang
charges #cnn	#ntc zintan	entreating super	##saad ##hariri	#d12 #ows
accusing sandusky	god particle	madonna halftime	##human ##rights	occupy wall
hearing today	fifth day	bowl halftime	#annahar economic	amazing content

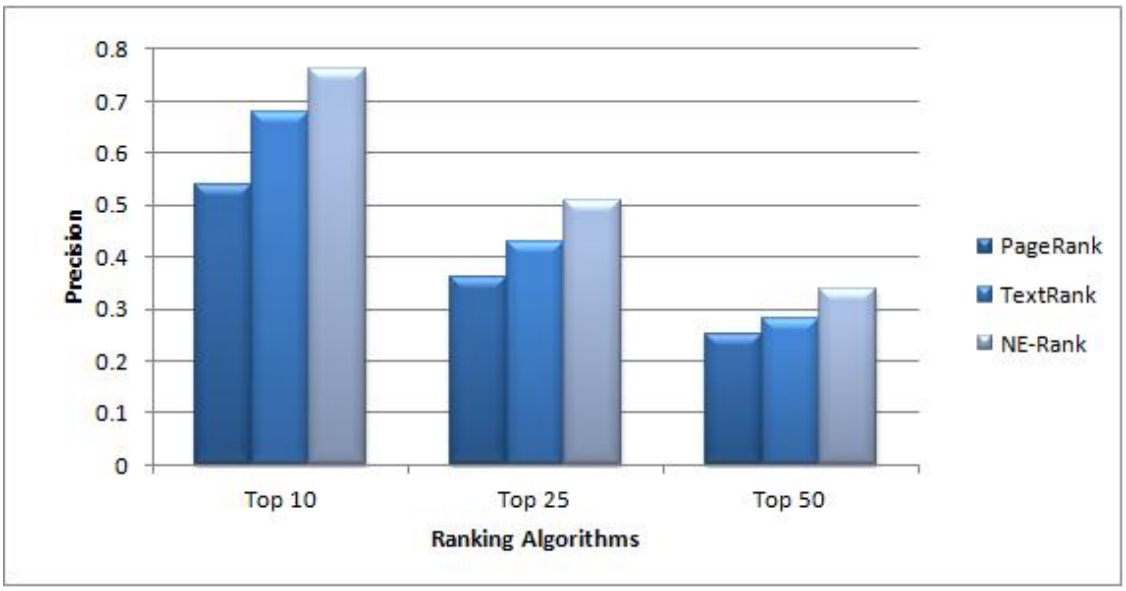


Figure 1. Comparison of Top 10, 25, and 50 for All Algorithms Using Precision

Table V
TOP 10 KEYPHRASES EVALUATION

	Precision	Bpref
NE-Rank	0.40	0.672
NE-Rank + Hashtags Boosting	0.48	0.654

keyphrases have improved the quality of the extracted keyphrases. Moreover, those keyphrases that are extracted of hashtag titles are very readable and understandable which is another great advantage of the approach. Even though a great improvement has been shown when hashtag title words are boosted especially with precision, it did not provide better results when using Bpref metric. The reason behind this is that a number of irrelevant hashtags were boosted regardless of our efforts in equation 5. For instance, the hashtag "#cameronlegacy" appearing in topic 10 which is about the Libyan revolution did not seem to be relevant for the evaluators; therefore, when it was boosted, it affected the order of the top 10. A qualitative evaluation of Table IV shows that keyphrases seem to be relevant to topics, informative, and clear. Extracted keyphrases could be used to summarize and represent tweets on the topics found. It could be useful for tracking topics and detecting events in a large corpus of tweets. Moreover, it could be used to identify topical terminologies that tend to occur with a specific topic more than others.

Different parameters settings can be very important in such research. Therefore, we have experimented with different parameters. The damping factor in the NE-Rank was tested with different values. We experimented with 0.10, 0.25, 0.50, and 0.85 with best results obtained with the latter. Moreover, the window size for identifying co-occurrence between words was tested with different values. Specifically, we checked the window size of 2, 5, and 10 with best results achieved when the window size is set to 2. The reason behind the window size of 2 having best results is the fact that tweets are very short documents.

In our future research, we plan to explore the following routes: To explore more advanced approaches for topic extraction as in Labeled LDA (L-LDA)[23] that uses more information to improve LDA. Moreover, we also plan to explore more approaches to improve the NE-Rank.

[4]"##sandusky ##hearing" is split from "#sanduskyhearing"

V. Conclusion

Summarizing the huge tweet collection and extracting topical keyprhases that would represent topics in Twitter is a challenging endeavor due to the massive amount, informality, noise, and short length of tweets. In this paper, we have used an extensive preprocessing approach to investigate such issues then introduced new features that might improve topical keyphrase extraction in Twitter. Consequently, we proposed a novel unsupervised graph-based keyword ranking method that considers words weight in addition to edges when calculating the ranking. Afterwards, we introduced a new approach of extracting keyphrases by leveraging hashtags. The potential and validity of both approaches have been demonstrated by conducting an experimental evaluation which produced an improvement when applying NE-Rank and hashtags enhanced extraction.

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